Juan de Dios Alvarez

Senior Data Scientist -- Applied AI, LLM Systems and Machine Learning for Financial Data

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Profile

Data scientist with seven years of industry work (2019 to present) on financial, regulatory and unstructured data, and a Master's in Computer Science (machine learning and NLP). At FIRA, a development bank, I built the pipeline behind its credit-assignment and non-payment-risk analysis. Since then I have built LLM and document-AI systems over bank statements, Mexican e-invoices (CFDI) and anti-money-laundering regulation, and I measure every model, retrieval method or prompt against a baseline or ground truth before it is relied on. I choose between generative AI and classical machine learning by measured accuracy, cost and runtime.

Selected Results

- **Generative AI, measured:** evaluated LLM extraction against a rule-based baseline (recall, false positives, tokens, runtime) and gated it on a labeled control set with a zero-missed-positives acceptance criterion.
- **Retrieval for regulation:** built keyword, embedding and hybrid (reciprocal rank fusion) retrieval over 26 Mexican AML legal instruments, compared by recall@k against a hand-labeled question bank.
- **Document AI on financial records:** cut doubtful lines from 175 to 77 on a fixed sample by comparing OCR, layout detectors and local vision-language models on character and word error rates.
- **Credit and risk data at scale:** replaced a single analyst's manual Excel process at FIRA with 74 scheduled extraction scripts (asynchronous Python) over 9+ sources feeding PostgreSQL and 15+ dashboards.

Professional Experience

Data Scientist & ML Engineer | Independent Practice

March 2026 -- present

- Designed LLM pipelines with mechanical guardrails (schema-constrained outputs, rejection of citations not in the source) and multi-model routing: a local LLM for bulk and sensitive reading, a frontier model for judgment, deterministic checks and human approval between steps; each call logs latency, tokens and truncation.
- Prototyped an AML compliance assistant (LFPIORPI): rules-as-data engine with rolling 6-month accumulation, UMA thresholds and filing deadlines, KYC flows, and the inputs and rule version stored behind every alert.
- Clustered short Spanish financial-transaction text without labels (multilingual embeddings, UMAP, HDBSCAN, FINCH), scored by silhouette, DBCV and ARI.
- Built and run in production a small business's data system: PostgreSQL warehouse (40+ tables), bank-to-invoice-to-payment reconciliation with confidence tiers, bank-statement OCR checked cell by cell against the PDF text layer, and a FastAPI application used daily by staff.

Economic Research Specialist (Data Scientist) | FIRA

January 2025 -- January 2026

- Replaced one analyst's manual Excel process, which fed credit-assignment and non-payment-risk analysis, with a scheduled, unattended pipeline: 74 extraction scripts (Python, aiohttp/asyncio, Playwright) over 9+ sources (INEGI, Banxico, USDA, others) loading PostgreSQL, with 60+ SQL functions and materialized views computing indicators.
- Built a crop-production model from Sentinel-2 and Planet Labs satellite vegetation indices: outlier filtering, PCA, Random Forest and Gradient Boosting selected by 5-fold cross-validated RMSE, evaluated on held-out data.
- Delivered automated executive reports and 15+ Apache Superset dashboards to decision-makers; set governance conventions (raw data preserved, lineage, GitLab) and supervised an undergraduate team.

Data Scientist | CITEIM

October 2024 -- January 2025

- Built the dataset for a Mexican Sign Language recognition system from video: asynchronous, rate-limited bulk download, weak labels from burned-in subtitles (OCR) and audio (offline speech-to-text), and MediaPipe pose and hand keypoints; designed a two-stage on-device pipeline.

Data Scientist & Team Lead | Punto Singular

December 2023 -- August 2024

- Led monthly analytics cycles for an EdTech client (Tableau, Looker); prototyped real-time WebSocket market-data processing on AWS EC2; founded and led a student Data Science group.

Data Scientist | BairesDev

September 2023 -- November 2023

- Built a retrieval-augmented generation proof of concept that let a legal client's lawyers query their own contracts for obligations, risk clauses and precedents; delivered transformer-based summarization and Neo4j graph modeling.

Data Scientist | D3Clarity (formerly Viqtor Davis North America)

February 2019 -- March 2023

- First data scientist of the company; defined the analytics roadmap for US-based clients in a distributed team.
- Took an NLP classifier of public school-board minutes from proof of concept to production on Azure ML, feeding a Power BI dashboard for the client's sales team; built forecasting models, data lakes and ETL on AWS.

Earlier Experience

2002 -- 2019

- Developer and services administrator, University of Michoacan IT (2013 -- 2019); lecturer, Tecnologico de Monterrey Business School; HAWC observatory data analysis (Python, C++); data-acquisition software, CERN Crystal Clear Collaboration (2002).

Education

- **Master's in Computer Science** -- INAOE, Mexico (2006 -- 2009). Degree awarded; SEP professional license (cedula) 5916605. Thesis: automatic text classification using prototype-based class reduction.
- **PhD Candidate in Physics** -- University of Michoacan (2009 -- 2013). HAWC data analysis; co-author on 14 publications.
- **Bachelor's in Physics and Mathematics** -- University of Michoacan (1998 -- 2003).
- **Certifications:** AWS Certified Cloud Practitioner; Semarchy xDM. **Languages:** Spanish (native), English (professional), Italian (basic-intermediate), French (basic).

Technical Skills

- **Generative AI:** RAG, embeddings (bge-m3, multilingual-e5, sentence-transformers), hybrid retrieval (BM25 + embeddings, RRF), multi-step LLM chains, schema-constrained extraction, source-grounding checks, local LLMs (Ollama), vision-language models, NER (GLiNER)
- **Evaluation and observability:** baselines and control sets, recall@k, CER/WER, k-fold cross-validation, RMSE/MAE/R2, silhouette/DBCV/ARI, KL divergence, stratified sampling; latency, token and truncation telemetry
- **Machine learning and statistics:** scikit-learn (Random Forest, Gradient Boosting, PCA), pandas, Polars, NumPy, SciPy, clustering (HDBSCAN, FINCH, UMAP); R; PyTorch at prototype level
- **Python engineering:** asyncio/aiohttp concurrency, FastAPI, Flask, pytest, modular pipelines, uv-locked dependencies, Git (GitHub, GitLab); AWS (EC2, SageMaker), Azure ML, Linux/systemd
- **Data:** PostgreSQL (PL/pgSQL, window functions, materialized views, JSONB), SQLite, Oracle, MySQL, Neo4j; ETL, lineage; text, scanned documents, satellite imagery, audio and video